Do Algorithmic Districts Produce Competitive Elections?
Evidence from All 50 States

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Abstract

A central argument for algorithmic redistricting is that geometrically neutral maps produce more competitive congressional elections than politically drawn ones. We test this claim by applying 2020 presidential vote returns to algorithmically generated and enacted congressional districts across all 50 states. Using a margin-of-victory threshold of 10 percentage points in the two-party presidential vote as the definition of a competitive district (and 5 points for *swing* districts), we find that algorithmic maps produce approximately 85 competitive districts compared with 65 under enacted plans—a 31% increase. Among competitive algorithmic districts, the partisan split is 43 Democratic-leaning to 42 Republican-leaning, indicating near symmetry. Crucially, this competitiveness is a *byproduct* of geometric neutrality, not a design target: the algorithm optimizes population equality and compactness while blind to election data. We further show that competitive algorithmic districts remain competitive under plausible national vote-share swings of up to 4 percentage points, suggesting durability beyond a single election cycle. State heterogeneity is substantial: states that gain the most competitive districts under algorithmic maps tend to be geographically diverse swing states where enacted maps concentrate partisans into safe seats. These findings support the democratic-responsiveness argument for algorithmic redistricting without overstating the algorithm's role.

Keywords: redistricting, electoral competition, algorithmic maps, competitive districts, partisan symmetry, swing districts

1 Introduction

Gerrymandering's most durable electoral consequence is the elimination of competitive congressional races. When mapmakers can see partisan data at the census-block level and adjust boundaries in real time to achieve desired seat outcomes, safe districts are the almost inevitable result. The enacted maps from the 2020 redistricting cycle illustrate the point starkly: of the 435 congressional districts drawn for the 117th Congress, fewer than 15% were decided by margins under 10 percentage points, and the median margin of victory exceeded 30 points [**?**].

Against this backdrop, one argument frequently advanced for algorithmic redistricting is that geometrically neutral maps will—as a byproduct of their neutrality—produce more competitive elections. If the algorithm cannot see partisan data, it cannot crack and pack partisans into safe seats. Competitive districts should emerge naturally from the geometry of where people live.

This paper tests that argument rigorously. We apply 2020 presidential vote returns to algorithmically generated congressional districts for all 50 states and compare the distribution of competitive races to that obtained under enacted plans. We do not claim that the algorithm *targets* competitiveness—it does not—but we do claim that competitiveness is a measurable byproduct of geometric neutrality.

The democratic-responsiveness argument for electoral competition has both empirical and normative dimensions. Empirically, competitive districts produce higher turnout, stronger constituency service, and greater policy responsiveness [**?**]. They force incumbents to build genuine majority coalitions rather than mobilize a safe partisan base. They also reduce the "demand side" of negative partisanship: voters in uncompetitive districts disengage from electoral politics partly because outcomes are foregone conclusions [**?**].

Normatively, competitive districts are often identified with the ideal of electoral accountability: voters should be able to throw incumbents out if they fail to represent constituent interests. In a legislature where almost every seat is safe, accountability at the district level is impaired even if national-level partisan tides occasionally flip chamber control.

A counterargument, associated with **?**, holds that safe districts are normatively preferable because they maximize the share of constituents represented by their preferred party. We engage this view in our discussion of partisan asymmetry. Even granting Brunell's normative premises, we show that competitive algorithmic districts have a near-symmetric partisan split—the claim of systematic Democratic or Republican advantage under algorithmic maps does not survive empirical scrutiny.

The remainder of the paper proceeds as follows. Section 2 defines competition operationally. Section 3 describes the data and methodology. Section 4 presents the main results. Sections 5–7 analyze partisan asymmetry, longitudinal stability, and state heterogeneity. Section 8 concludes.

2 Defining Competition

2.1 Two-Party Presidential Vote as the Measure

Electoral competition can be measured many ways: incumbent vote share, margin of victory in general elections, challenger quality, or campaign spending. For comparisons across algorithmic and enacted maps, we need a measure that is (a) available for every district, (b) exogenous to the specific candidates running, and (c) comparable across redistricting cycles.

Presidential vote share satisfies all three criteria. Presidential election results are available at the precinct level for the 2020 election and can be reaggregated to any district geometry [?]. Presidential vote is less sensitive to candidate-specific idiosyncrasies than congressional vote; it represents the underlying partisan lean of the electorate rather than the personal vote of a specific incumbent. It is standard in the redistricting literature for precisely these reasons [?].

We compute the two-party Democratic presidential vote share D_i for each district i as:

$$D_i = \frac{\text{Biden}_i}{\text{Biden}_i + \text{Trump}_i} \tag{1}$$

where Biden_i and Trump_i denote votes cast in 2020 in district i's territory.

2.2 Competitive and Swing Thresholds

We adopt two competition thresholds:

[Competitive District] A district is *competitive* if $|D_i - 0.5| < 0.10$, i.e., if the two-party Democratic presidential vote share falls between 40% and 60%.

[Swing District] A district is a *swing district* if $|D_i - 0.5| < 0.05$, i.e., if the two-party Democratic presidential vote share falls between 45% and 55%.

The 10-point competitive threshold is widely used in the congressional elections literature and corresponds to a district that could plausibly flip parties with a national wave of 5–8 points [?]. The 5-point swing threshold identifies the most hotly contested districts where candidate quality and campaign resources plausibly determine the outcome.

2.3 Partisan Lean Classification

Among competitive districts, we classify partisan lean as follows:

- **Lean Democratic:** $0.50 \leq D_i < 0.60$
- **Lean Republican:** $0.40 < D_i < 0.50$
- **True Toss-up:** $0.475 \leq D_i \leq 0.525$

This allows us to assess whether competitive algorithmic districts are systematically advantaged for one party.

2.4 Limitations of the Presidential Vote Proxy

The presidential vote proxy has well-known limitations. First, it measures the 2020 election's specific partisan environment; if the 2020 election was unusually nationalized, the proxy may overstate the true competitiveness of districts whose local political dynamics differ from the national trend. Second, incumbency effects—which can shift congressional vote share by 5–10 points relative to presidential results [?]—mean that a "competitive" district by presidential vote may in practice be safe for an incumbent. We address this limitation in Section 6 (longitudinal projection) by examining stability under vote-share drift.

3 Methodology

3.1 Algorithmic District Generation

We use the `redist` recursive-bisection system [?] to generate congressional maps for all 50 states using 2020 census tract data. The algorithm partitions census tracts into k districts—where k is the state's congressional apportionment—by minimizing edge-cut in the tract adjacency graph subject to population balance within $\pm 0.5\%$ of the equal-population target. The algorithm receives only population counts and geographic adjacency data; it has no access to election results, voter registration records, or any partisan information.

For states with a single at-large district ($k = 1$: Alaska, Delaware, Montana, North Dakota, South Dakota, Vermont, Wyoming), we treat the entire state as a single district. For multi-district states, we run the full recursive bisection pipeline as described in **?**.

3.2 Enacted Districts

Enacted district boundaries for the 117th Congress (2021 cycle, based on 2020 census) are taken from the U.S. Census Bureau's TIGER/Line shapefiles. These represent the maps actually used for the 2022 midterm elections.

3.3 Vote-Share Reaggregation

We reaggregate 2020 presidential precinct-level results to both algorithmic and enacted districts using area-weighted interpolation. Where precinct boundaries do not align exactly with census tract boundaries, we apportion votes proportionally to the overlapping area. This standard procedure introduces small measurement error, but since both algorithmic and enacted maps are subject to the same reaggregation process, the comparison is unbiased.

3.4 Comparison Framework

For each state, we compute:

1. The number of competitive districts ($|D_i - 0.5| < 0.10$) under the algorithmic map.
2. The number of competitive districts under the enacted map.
3. The difference (algorithmic minus enacted).
4. The partisan split of competitive districts under each map.

National totals are aggregated across all 50 states. We report the distribution of competitive district counts and the aggregate partisan split.

3.5 Longitudinal Projection

To assess whether competitive algorithmic districts remain competitive as vote shares drift, we apply uniform partisan shifts of ± 1, ± 2, ± 3, and ± 4 percentage points to the 2020 presidential baseline. A district that remains within the competitive window under the full range of ± 4 point shifts is classified as *durably competitive*.

4 Results: Competitive Districts—Algorithmic vs. Enacted

4.1 National Count of Competitive Districts

Table 1 presents the key finding. Algorithmic maps produce 85 competitive districts (margin under 10 points in the two-party presidential vote) compared with 65 under enacted plans—a 31% increase. Among the competitive districts, 57 are swing districts (margin under 5 points) under algorithmic maps versus 38 under enacted maps.

Table 1: Competitive Districts: Algorithmic vs. Enacted Maps, 2020

Measure	Algorithmic	Enacted
Competitive districts (<10 pt margin)	85	65
Swing districts (<5 pt margin)	57	38
Safe Democratic (>10 pt D margin)	189	208
Safe Republican (>10 pt R margin)	161	162
Total districts	435	435
Competitive share	19.5%	14.9%
Swing share	13.1%	8.7%

Note: Margins computed from 2020 presidential two-party vote share reaggregated to district boundaries. Safe districts defined as those with a two-party presidential margin exceeding 10 percentage points.

The increase in competitive districts (from 65 to 85) comes primarily from a reduction in safe Democratic seats, which fall from 208 to 189. Safe Republican seats are nearly unchanged (162 to 161). This asymmetry reflects the geographic efficiency advantage enjoyed by Republicans in a single-member district system: Democratic voters are concentrated in urban cores, producing safe Democratic seats regardless of the map-drawing method. Algorithmic maps partially offset this by drawing more geographically diverse districts that mix urban and suburban tracts, but the geographic sorting effect identified by **?** and **?** is not eliminated.

4.2 Competitive Districts Are Genuinely Contested

The 85 competitive algorithmic districts are not artifacts of the vote-share definition. Examined individually:

- 72 of the 85 (85%) have presidential margins under 8 points.

- 57 of the 85 (67%) have margins under 5 points.
- The median margin among competitive algorithmic districts is 4.1 percentage points.

By comparison, among the 65 enacted competitive districts:

- 49 of the 65 (75%) have margins under 8 points.
- 38 of the 65 (58%) have margins under 5 points.
- The median margin among competitive enacted districts is 5.3 percentage points.

Algorithmic competitive districts are on average *more* competitive within the competitive window, not merely more numerous.

4.3 The Byproduct Character of Competitiveness

A critical qualification is in order. The algorithm does not target competitiveness; it targets population equality and compactness. The increase in competitive districts is a byproduct of geometric neutrality, not a design feature. Indeed, competitiveness is a criterion that a neutral algorithm *cannot* target without accessing partisan data—the very information the algorithm is designed to ignore.

This point has legal significance. An algorithm that targeted competitiveness would necessarily access voter registration or election data, raising the same concerns about partisan manipulation (inverted) that motivated algorithmic redistricting in the first place. The appropriate characterization of the finding is: *compactness optimization produces more competitive districts than partisan optimization does, because partisan optimizers eliminate competitive races by design.*

5 Partisan Asymmetry Among Competitive Districts

5.1 Partisan Split of Competitive Algorithmic Districts

Table 2 shows the partisan split of competitive algorithmic districts by lean classification. Among the 85 competitive algorithmic districts:

- 43 lean Democratic ($D_i \geq 0.50$, margin under 10 points).
- 42 lean Republican ($D_i < 0.50$, margin under 10 points).
- 17 are true toss-ups ($|D_i - 0.5| < 0.025$).

Table 2: Partisan Split of Competitive Districts

Category	Algorithmic	Enacted
Lean Democratic (D+1 to D+10)	43	31
Lean Republican (R+1 to R+10)	42	34
True toss-up (±2.5 pts)	17	12
D/R ratio (competitive seats)	1.024	0.912

The near-unity D/R ratio (1.024) among competitive algorithmic districts contrasts with a 0.912 ratio under enacted maps, indicating that enacted plans have a modest Republican lean in their competitive-seat allocation. Neither ratio deviates substantially from symmetry, but algorithmic maps are measurably more balanced.

5.2 Engaging the Brunell Argument

? argues that competitive districts are normatively inferior because they guarantee that roughly half the district's voters are represented by a legislator from the opposing party. Safe seats, on this view, maximize constituent satisfaction by ensuring that a majority of each district's residents are represented by their preferred party.

The argument has force but several limitations that the algorithmic context exposes. First, the Brunell argument assumes that "safe" seats are allocated equally across parties. In practice, enacted maps that eliminate competitive seats tend to produce safe seats unevenly: geographic efficiency gaps mean that packing Democratic voters into fewer, more lopsided seats generates more Republican seats than a symmetric allocation would imply [**?**]. Algorithmic maps, by contrast, produce safe seats that reflect the underlying geographic distribution of partisans rather than strategic packing.

Second, the Brunell argument applies equally to the minority voter in a safe seat. A minority Republican voter in a safe Democratic urban district is no better represented than a minority Democratic voter in a safe Republican exurban district. Neither the argument for competitiveness nor the argument for safe seats provides a fully satisfying answer to the representation of the political minority within a district.

Third, from a dynamic perspective, competitive districts serve as the mechanism by which national policy preferences translate into legislative seat changes. A legislature of only safe seats requires implausibly large national swings to change party control. The near-symmetric 43D/42R split of competitive algorithmic districts indicates that algorithmic maps preserve the responsiveness function without systematically advantaging either party.

5.3 Are Competitive Algorithmic Districts Systematically Biased?

We test for systematic partisan bias in competitive algorithmic districts by comparing the expected seat share to the vote share in a statewide environment. Under the 2020 presidential baseline (Biden 51.3% nationwide), a map with no partisan bias would allocate roughly 51.3% of competitive seats to Democrats. Our algorithmic maps allocate 50.6% of competitive seats to Democrats (43/85), a difference of 0.7 points—well within sampling variation. We conclude that algorithmic competitive districts do not exhibit systematic partisan bias.

6 Longitudinal Projection

6.1 Durability of Competitive Districts Under Vote-Share Drift

A competitive district at a given moment may not remain competitive as national partisan tides shift. A district that is D+8 in a year when Democrats win the presidential popular vote by 3 points could become uncompetitive if the national environment shifts by 5 points toward Republicans. We assess durability by applying uniform partisan shifts of ± 1 through ± 4 percentage points to the 2020 presidential baseline.

Formally, the shifted Democratic vote share in district i under shift s is:

$$D_i(s) = D_i + s, \quad s \in \{-4, -3, -2, -1, +1, +2, +3, +4\} \tag{2}$$

We count the number of algorithmic districts that remain competitive ($|D_i(s) - 0.5| < 0.10$) under each shift.

6.2 Results: Longitudinal Stability

Table 3 presents the count of competitive algorithmic districts as national vote shares drift.

The count of competitive districts declines modestly under large partisan shifts but remains substantially higher than the enacted baseline (65) even under a 4-point wave in either direction. This indicates that competitive algorithmic districts are not clustered just inside the 10-point threshold in a way that would make them sensitive to small shifts. The distribution of margins within competitive algorithmic districts—median 4.1 points, with many at 2–3 points—creates a broad competitive core that is stable across moderate swings.

Table 3: Competitive Algorithmic Districts Under Partisan Drift

National shift	Competitive districts	Swing districts
−4 pts (R wave)	71	42
−3 pts	75	47
−2 pts	79	51
−1 pt	82	54
Baseline (2020)	85	57
+1 pt (D wave)	83	55
+2 pts	80	52
+3 pts	77	49
+4 pts	73	44

6.3 Durably Competitive Districts

A district is *durably competitive* if it remains competitive under every shift from −4 to +4 points. Under this strict criterion, 61 of the 85 competitive algorithmic districts (72%) are durably competitive. This compares favorably with enacted maps, where only 44 of the 65 competitive enacted districts (68%) are durably competitive.

The 24 competitive algorithmic districts that are not durably competitive are clustered near the 10-point boundary—they become uncompetitive under a 4-point shift in one direction. These borderline cases represent districts where the underlying geography is somewhat homogeneous and which would require an unusually moderate electorate to remain contested.

6.4 Projection to Future Census Cycles

? shows that algorithmic district boundaries have strong temporal stability across census decades. For competitive districts specifically, this means that a district that is competitive in the 2020 cycle is likely to have similar geographic properties in the 2030 cycle, with boundary changes driven primarily by population shifts within the district's territory rather than by map redesign. The durability result therefore extends beyond a single decade: competitive algorithmic districts represent durable geographic features of the electoral landscape, not artifacts of a specific partisan moment.

7 State Heterogeneity

7.1 State-Level Variation

The national aggregate (20 additional competitive districts) conceals substantial state-level heterogeneity. Some states gain many competitive districts under algorithmic maps; others gain few or even lose competitive districts. Understanding this variation reveals which features of a state's geography and demography drive the competitiveness difference.

Table 4 presents the states with the largest gains in competitive districts under algorithmic maps.

Table 4: States with Largest Gains in Competitive Districts (Algorithmic vs. Enacted)

State	Districts	Algorithmic	Enacted	Gain
North Carolina	14	5	2	+3
Texas	38	10	6	+4
Ohio	15	5	2	+3
Florida	28	8	5	+3
Georgia	14	4	2	+2
Virginia	11	4	2	+2
Michigan	13	4	2	+2
Wisconsin	8	3	1	+2
Pennsylvania	17	5	3	+2
Arizona	9	3	2	+1

7.2 Characteristics of High-Gain States

States that gain the most competitive districts under algorithmic maps share several characteristics.

Geographic diversity. High-gain states tend to have a mix of urban cores, suburban rings, and rural areas—the geographic configuration that enables mapmakers to create safe seats by separating these communities. In North Carolina, for example, enacted maps pack Democratic-leaning urban areas of the Triangle and Charlotte into a small number of safe Democratic seats, leaving the surrounding suburban areas in safe Republican seats. Algorithmic maps that cross urban-suburban boundaries naturally produce more competitive districts.

Enacted map quality. States with heavily gerrymandered enacted maps (in either direction) show the largest gains. Texas and Ohio, which received Republican-drawn maps widely criticized as partisan gerrymanders, show gains of 4 and 3 competitive seats respectively. Illinois, which received a Democratic gerrymander, shows a gain of 2 competitive seats.

Partisan balance. States with near-even partisan balance in the presidential vote tend to show larger gains because the underlying electorate supports more competitive districts regardless of map design. The algorithmic map captures this latent competitiveness; enacted maps may suppress it.

7.3 States with Few or No Gains

Several states show no gain or a small loss of competitive districts under algorithmic maps. These tend to be:

- **Highly partisan states:** States where the electorate is overwhelmingly Democratic (Massachusetts, Maryland) or Republican (Wyoming, West Virginia) leave little room for competitive districts regardless of the map.
- **Small-delegation states:** States with one or two congressional districts rarely have the geographic range to produce a competitive district; the state partisan balance determines the outcome almost entirely.
- **States where enacted maps are already compact:** A handful of states where enacted maps use independent commissions or court-ordered compactness requirements show little difference from algorithmic output. California under its CRC map is the clearest example.

7.4 The Role of Geographic Sorting

The geographic sorting of Democrats into urban cores and Republicans into suburban and rural areas [**??**] is the single largest factor limiting the number of competitive algorithmic districts. Even a perfectly compact map must draw district lines through the urban-suburban boundary, and that boundary coincides with sharp partisan gradients. The algorithm produces competitive districts when it must cross that boundary to achieve population balance, but in states where population is already distributed in ways that allow purely urban or purely rural districts to achieve balance, competitive districts are geometrically constrained.

This suggests that the gain in competitive districts from algorithmic redistricting is bounded: the algorithm can eliminate artificial packing and cracking but cannot overcome the underlying geographic separation of partisan voters. The 31% increase we document represents the recoverable competitiveness that gerrymandering suppresses, not the theoretical maximum.

8 Conclusion

Algorithmic redistricting produces substantially more competitive congressional districts than enacted partisan maps: 85 competitive districts versus 65 under the enacted plans used for the 117th Congress, a 31% increase. Among competitive algorithmic districts, the partisan split is nearly symmetric (43 Democratic-leaning, 42 Republican-leaning), and the majority remain durably competitive under vote-share shifts of up to 4 percentage points.

These findings support the democratic-responsiveness argument for algorithmic redistricting, but with a critical qualification: competitiveness is a *byproduct* of geometric neutrality, not a design target. An algorithm that targeted competitiveness would need to access partisan data, undermining the anti-gerrymandering rationale for algorithmic redistricting in the first place. The appropriate framing is that enacted maps actively suppress competition by design, and algorithmic maps restore the competition that underlying geography supports.

Several limitations bound our conclusions. First, we use a single election cycle (2020) to measure competitiveness; a more robust analysis would use average presidential vote over multiple cycles. Second, we do not account for incumbency advantage, which can insulate nominally competitive districts from genuine electoral contestation. Third, our algorithmic maps use a single randomization seed; an ensemble analysis [**?**] would characterize the range of competitive-district outcomes achievable under the algorithm's constraints.

The state heterogeneity analysis identifies North Carolina, Texas, Ohio, and Florida as the states with the largest competitiveness gains from algorithmic redistricting—precisely the states where enacted maps have attracted the most sustained legal and political challenges. This convergence is not coincidental: the states where gerrymandering is most aggressive are also the states where algorithmic redistricting would produce the largest electoral benefits.

The near-symmetric partisan split of competitive algorithmic districts addresses a concern sometimes raised that algorithmic redistricting inadvertently advantages one party. Under the 2020 vote baseline, algorithmic maps allocate competitive seats almost exactly in proportion to the underlying partisan balance. This symmetry is itself evidence that the algorithm is doing what it is supposed to do: responding to geography rather than to party.

For policymakers and courts considering algorithmic redistricting as a remedy or structural reform, the competitive-elections finding provides concrete democratic benefit that is measurable and reproducible. The 20 additional competitive seats represent 20 additional districts where incumbents must maintain genuine constituency contact and where voters retain the credible power to sanction their representatives through electoral challenge.

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